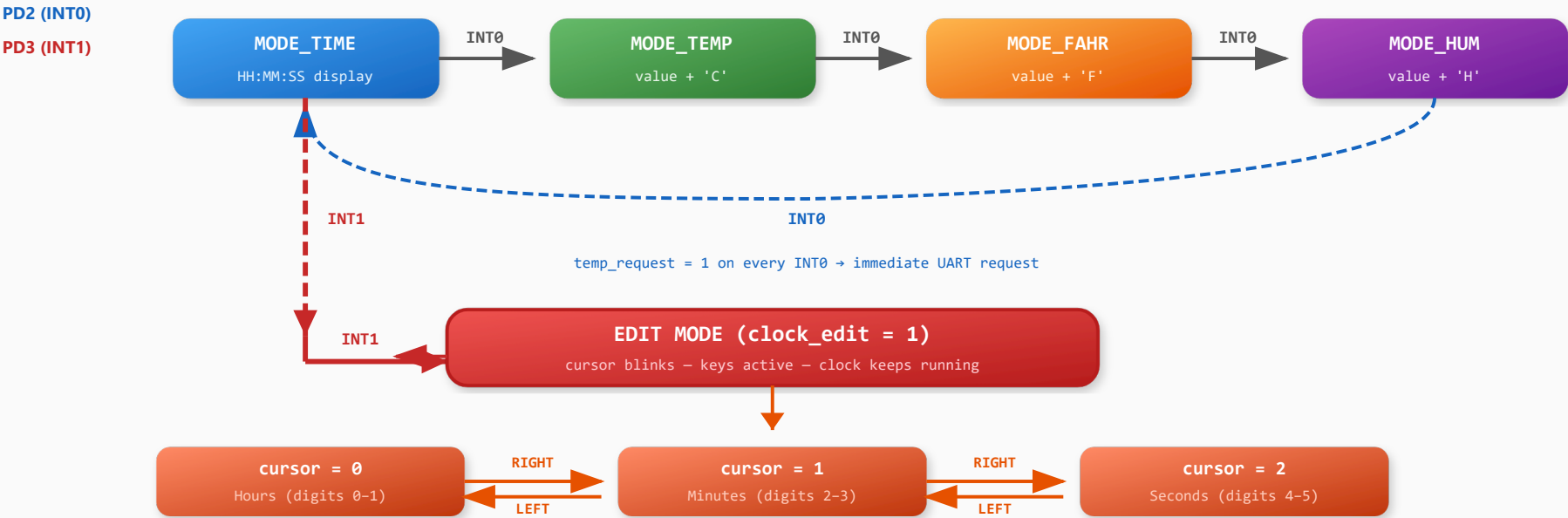


ATmega32 — FSM / Workflow Diagram



Key Actions (only when clock_edit = 1)

UP (PA0) – Increment

cursor=0: hours++ (wrap 0→23→0) | cursor=1: minutes++ (0→59→0)
cursor=2: seconds++ (wrap 0→59→0)

DOWN (PA1) – Decrement

cursor=0: hours-- (wrap 0→23) | cursor=1: minutes-- (0→59)
cursor=2: seconds-- (wrap 0→59)

B1 – Blinking Cursor

set_blink(i, on) | blink[i]=on | ISR: if (blink[cur] && (tick>>6)&1) skip digit

Legend

- Display mode
- Edit mode
- Cursor sub-state
- INT0 return
- INT1 toggle
- Key navigation

Triggers: INT0=PD2 (mode cycle) | INT1=PD3 (edit toggle) | Keys=PA0-3 (edit adjust)
Timers: T1=1s tick | T2=2ms display refresh, key scan, heartbeat, blink